

Ex.No: 8

Use Steg-Expose to Detect Hidden Data in Images

Date:-

Aim:-

To detect hidden data embedded in images using StegExpose by analyzing their statistical properties.

Description:-

StegExpose is a tool used to detect hidden data in images using steganography analysis. It works by evaluating the statistical properties of an image to estimate whether any hidden data is embedded.

Below is a step-by-step guide on how to use StegExpose to detect hidden data in images:

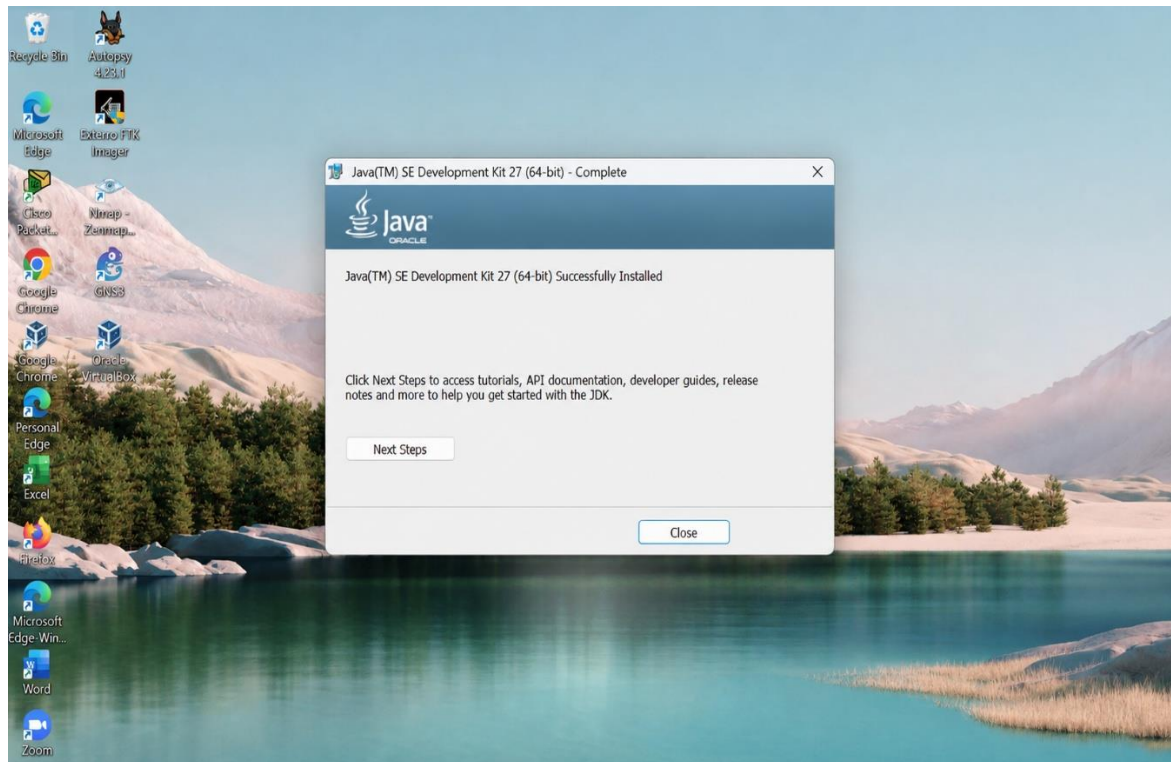
Prerequisites:

1. **Java Runtime Environment (JRE):** StegExpose is a Java-based tool, so ensure you have JRE installed.
2. **StegExpose Tool:** Download StegExpose from the [official GitHub repository](#).

Step-by-Step Process:

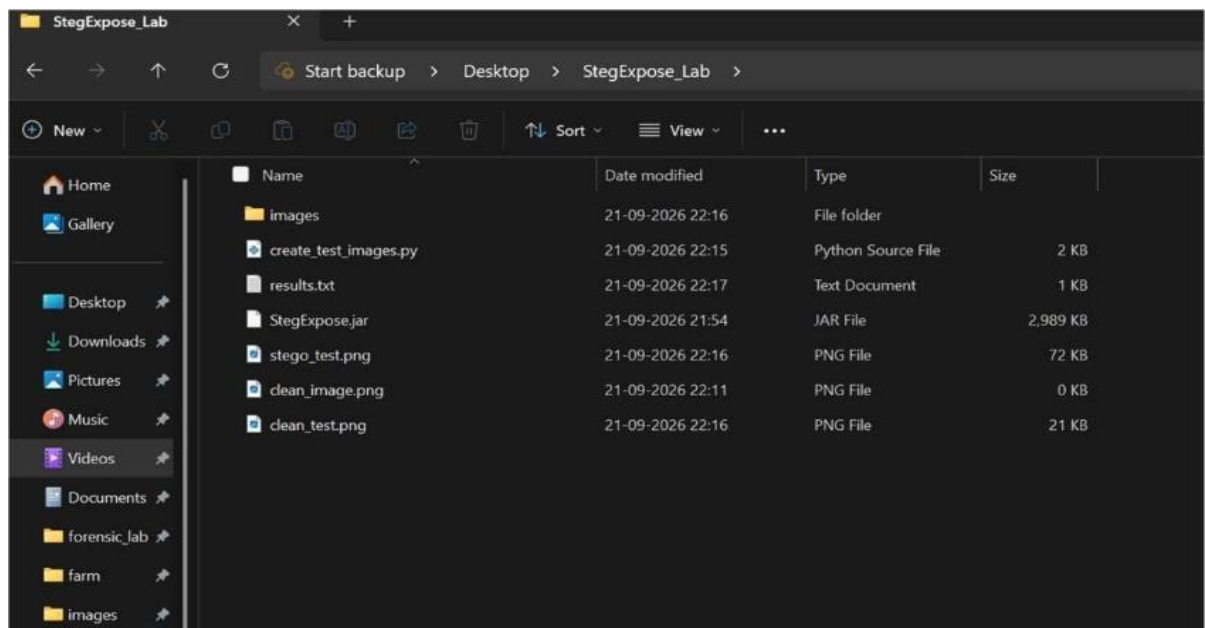
1. Download and Set Up StegExpose:

- **Download the tool:** Go to [StegExpose GitHub](#) and download the .jar file.
- **Install Java:** Ensure that Java is installed on your machine. You can download it from the official site.
- **Prepare environment:** Place the .jar file in a folder where you plan to work on the image files.



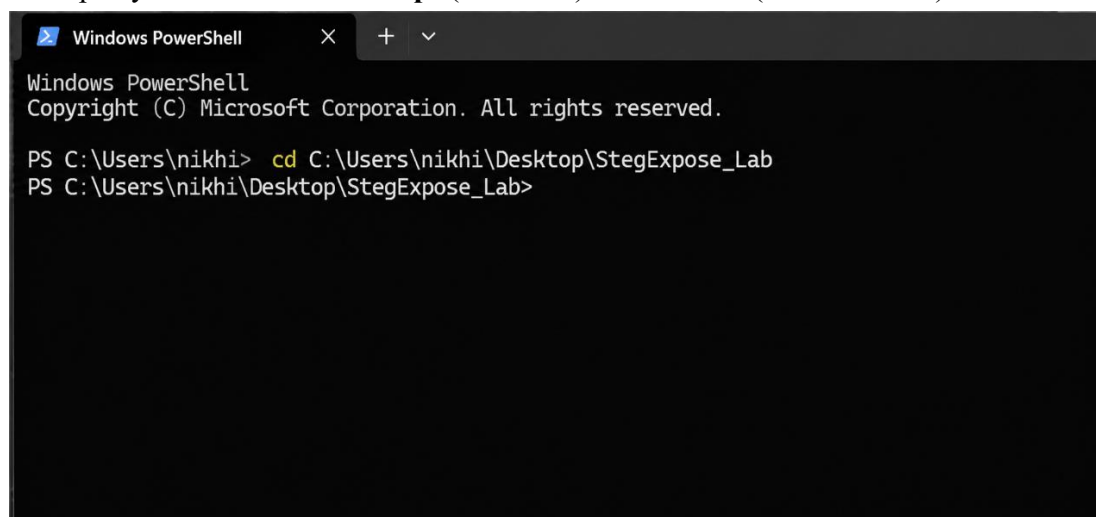
2. Select Images for Analysis:

- Collect the images you suspect might have hidden data. StegExpose supports formats such as .png, .jpg, .bmp, etc.



3. Open Command Line or Terminal:

- Navigate to the folder where you have saved the StegExpose.jar file.
- Open your **Command Prompt** (Windows) or **Terminal** (Linux/macOS).



4. Run StegExpose on an Image:

- Use the following command to check a single image for hidden data:

bash

java -jar StegExpose.jar <image_file_path> Example:

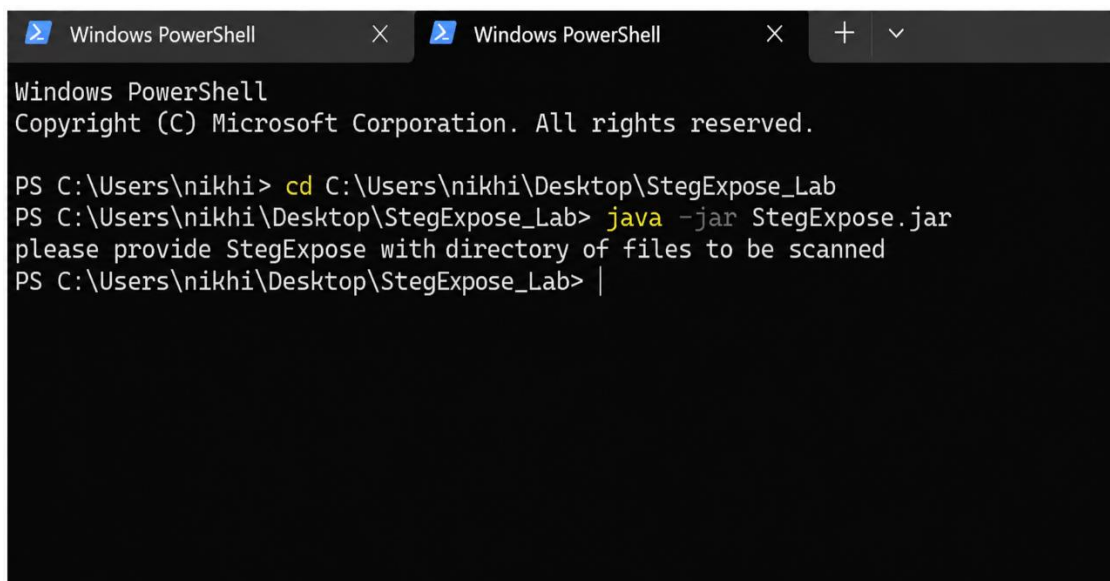
bash

java -jar StegExpose.jar test_image.png

- Replace <image_file_path> with the actual path of the image you want to check.

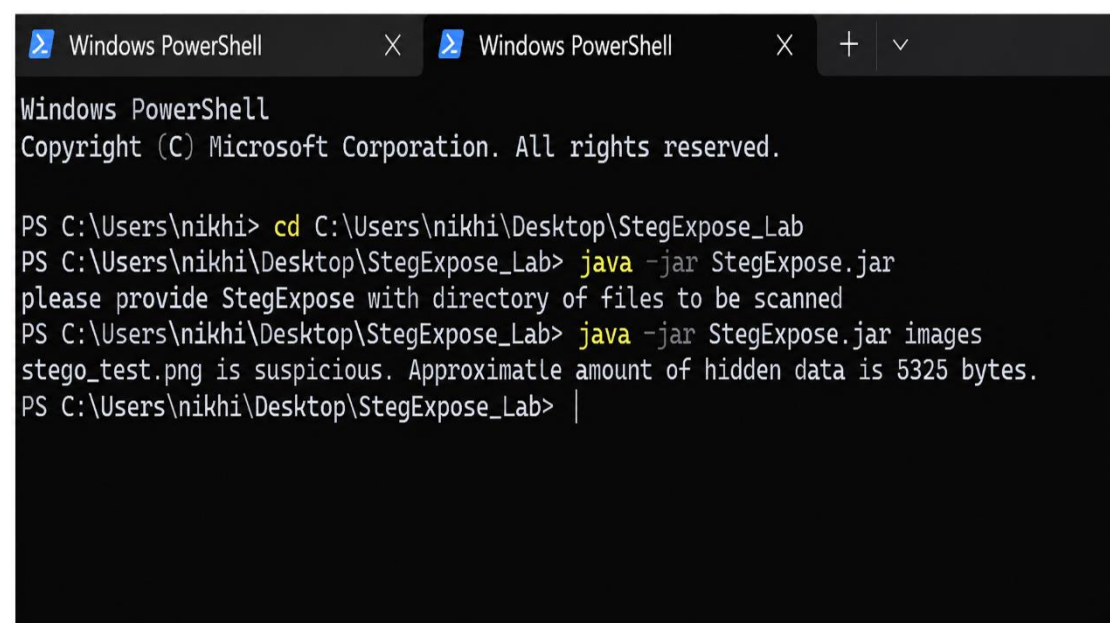
5. Analyze the Output:

- After running the tool, it will analyze the image and provide a score.
- **Output Explanation:**
 - StegExpose calculates a "suspect" score (between 0 and 1). The higher the score, the more likely that hidden data is present.
 - **Threshold values** (suggested by the tool):
 - **Less than 0.2:** Image is clean (no hidden data).
 - **0.2 - 0.3:** Possibly some hidden data.
 - **Above 0.3:** Likely that steganography is present.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\nikhi> cd C:\Users\nikhi\Desktop\StegExpose_Lab
PS C:\Users\nikhi\Desktop\StegExpose_Lab> java -jar StegExpose.jar
please provide StegExpose with directory of files to be scanned
PS C:\Users\nikhi\Desktop\StegExpose_Lab> |
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\nikhi> cd C:\Users\nikhi\Desktop\StegExpose_Lab
PS C:\Users\nikhi\Desktop\StegExpose_Lab> java -jar StegExpose.jar
please provide StegExpose with directory of files to be scanned
PS C:\Users\nikhi\Desktop\StegExpose_Lab> java -jar StegExpose.jar images
stego_test.png is suspicious. Approximate amount of hidden data is 5325 bytes.
PS C:\Users\nikhi\Desktop\StegExpose_Lab> |
```

6. Batch Analysis (Multiple Images):

- If you have multiple images to check, you can analyze them in bulk by specifying a folder. For example:

```
bash
```

```
java -jar StegExpose.jar
```

- Replace with the path to the folder containing the images.

7. Advanced Options (Optional):

- StegExpose provides additional parameters, such as adjusting the sensitivity of detection and output verbosity. Use --help to see the list of options:

```
bash
```

```
java -jar StegExpose.jar --help
```

8. Review the Results:

- For each image analyzed, review the scores and determine whether you need to investigate further.

Rubrics:-

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:-

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.